



Introduction to tuples

As a data professional, sometimes it will be more important to access and reference data than to change and manipulate it. When you simply need to find some information, but keep the data intact, you can use a data structure called a tuple. A tuple is an immutable sequence that can contain elements of any data type. Tuples are kind of like lists, but they're more secure because they cannot be changed as easily. They're helpful because they keep data that needs to be processed together in the same structure. Tuples are instantiated, or expressed, with parenthesis or the tuple() function. Here we have a tuple that represents someone's full name.

Notice that it was instantiated using parentheses. The first element of the tuple is their first name, the second is the first letter of their middle name, and the third is their last name. The position of the element is fixed in tuples, so you can't add new elements in the middle, and you can't change any of the elements. If we try to change the last name, which lives in index number 2, from Hopper to Copper, the code will throw an error. You can add a value to the end, but only if you reassign the tuple. Another way we can create tuples is by using the tuple function to transform input into tuples. In this case, our name is represented as a list.

We can convert the list to a tuple by using the tuple function. Notice that the name is no longer a list, so it doesn't have the brackets anymore. Tuples are also used to return values and functions. In fact, when a function returns more than one value, it's actually returning a tuple. For example, here's a function that takes as an argument a float value that represents a price. The function returns the number of dollars and cents. Notice that when we use the function to convert \$6 and 55 cents to dollars and cents, the return value is represented as a tuple that contains two numbers.

Interestingly, even though **tuple are immutable**, **they can be split into separate variables**.

When we run the "to dollar cents" function, we can directly assign the output into distinct variables. The information stored as a tuple in the result variable has now been reassigned to two separate variables that we can manipulate as we please. This process is known as unpacking a tuple. Notice that the unpacked variables themselves are no longer tuples. In this case, they're integers. A big advantage of working with tuples is that they let you store data of different types inside other data structures.

Here's an example of how this might be useful. This is a list of the starting five players on a university women's basketball team. Each player is represented by a tuple that contains their name, age, and position. This is a useful way of working with this type of information. The order of the players doesn't matter that much, and we might want to add to or rearrange them. So we use a list, which is mutable. However, the players themselves are individual records that are represented by tuples.

They are a bit more secure because tuples are immutable and more difficult to accidentally change. Because lists and tuples are iterable, we can extract information from them using loops. For example, we can write a "for" loop that unpacks each tuple

into three separate variables, and then print one of the variables for each iteration. This is equivalent to looping over each player record and printing the record at index zero. Using tuples in data professional work helps make your processes more efficient. It saves memory and can really optimize your programs too. Plus, when others collaborate with you on your code, your use of tuples will make it clear to them that your sequences of values are not intended to be modified. This is yet another great way to save your team time and effort.

Tuple

An immutable sequence that can contain elements of any data type

Tuples are expressed with parentheses or the `tuple()` function.

`tuple()`

Function that transforms input into tuples

Question



A tuple is an immutable sequence that can contain elements of any data type.

☒ True

☐ False

✓ **Correct**

A tuple is an immutable sequence that can contain elements of any data type.

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